JIAYI (MAGGIE) SHAO

Research Engineer | Iyer Lab, University of Washington

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Education

University of Washington Sep 2024 – Jun 2026
M.S., Electrical and Computer Engineering
GPA: 3.82/4.00

Shanghai University Sep 2019 – Jun 2023
B.E., Electrical Engineering and Automation
GPA: 3.62/4.00 | Top 5%

Research Experience

GM Post-Crash Sensing System Jul 2026 – Present
Project Lead; Advisor: Prof. Vikram Iyer

- Led development of a distributed post-crash sensing system with General Motors that detects collisions, collects crash information, identifies vehicle occupancy and cries for help using on-device ML, and transmits crash reports via LoRa
- Fine-tuned camera and audio models for on-device occupant detection and cries-for-help recognition, and benchmarked their accuracy and robustness
- Designed custom PCBs integrating motion sensors, camera, audio, wireless communication, power management, and RF wireless charging for distributed in-vehicle sensing
- Designed a low-power sensing system that wakes on vehicle motion and powers on the camera and microphone only after a collision, achieving 121 μA current while parked
- Developed an RF energy harvesting and wireless charging system, achieving up to 532 μA charging current in vehicle tests and enabling energy-neutral operation while parked

NailPose May 2026 – Sep 2026
Research Contributor; Advisor: Prof. Qiuyue (Shirley) Xue

- Developed NailPose, a distributed wireless smart nail system that reconstructs 3D whole-hand pose from four nail-mounted accelerometers, achieving 8.91 mm mean aligned reconstruction error
- Designed a 10 × 9 mm flexible PCB integrating sensing and wireless electronics into a 0.65 g nail sticker, and developed a secure nail-attachment method
- Developed embedded firmware on the nRF52832 for 100 Hz accelerometer sampling, efficient data buffering, and wireless data transmission over BLE
- Implemented an accelerometer-based double-tap wake-up mechanism and low-power standby mode, achieving 27.52 μA standby current and an estimated 14.2-day battery life with intermittent use

E-Textile Yarn for Continuous Auscultation Jan 2026 – Present
Project Lead; Advisors: Prof. Tingyu Cheng and Prof. Shwetak Patel

- Led development of an e-textile yarn with embedded contact microphones for continuous, 24/7 chest-worn auscultation
- Designed and fabricated miniaturized flexible PCBs, integrated them into a functional e-textile yarn, and developed a wireless audio acquisition system

· Evaluated spectral similarity of raw audio recordings against a reference stethoscope, measuring a pre-denoising spectral MAE of 8.69 dB

Multi-site PPG Nov 2025 – May 2026
Project Lead; Advisor: Prof. Qiuyue (Shirley) Xue

· Led Multi-site PPG, a large-scale in-the-wild physiological sensing dataset with 350+ hours of recordings per device across four custom wearables (earring, ring, watch, and necklace)
· Designed and conducted multi-day, in-the-wild user studies with 20 participants, developing user-friendly study protocols for reliable data collection
· Improved wearable usability and robustness through hardware prototyping, including custom 3D-printed housings, removable battery connectors, charging solutions, and protective silicone cases
· Developed data processing pipelines for multi-device synchronization, quality control, and heuristic heart-rate estimation to support downstream HR/HRV benchmarking

Embedded Arena Sep 2025 – Nov 2025
Research Contributor; Advisors: Prof. Vikram Iyer and Prof. Shwetak Patel

· Contributed to Embedded Arena, a hardware-in-the-loop framework where LLM agents optimize on-device ML models using performance measurements from real microcontrollers
· Customized PCB designs and contributed embedded firmware to support hardware-in-the-loop model evaluation

Employment

University of Washington – Iyer Lab Jul 2026 – Present
Research Engineer; Advised by Prof. Vikram Iyer

· Conduct research on intelligent sensing systems, focusing on custom hardware, low-power embedded design, and on-device ML

MH3D Inc. Jun 2025 – Sep 2025
Hardware Design Intern

· Designed the AFE board's analog signal path for a multi-channel SPECT medical imaging readout system, filtering, amplifying, and converting HEXITEC ASIC outputs to differential signals for ADC sampling

Huatai Automation Technology Jun 2022 – Oct 2022
Power Electronics Development Intern

· Invented a method for detecting parasitic parameters in power circuits to guide snubber tuning for an industrial switching power supply (patented)
· Designed schematics and PCBs for three candidate solutions and cut high-frequency noise 10× (2 V to 200 mV) through power integrity analysis, passing EFT and ESD tests

Publications

1. **Multi-site PPG: An In-the-Wild Physiological Dataset from Emerging Multi-site Wearables**
Conference on Neural Information Processing Systems (NeurIPS), Evaluations & Datasets Track, 2026

Jiayi Shao, Jiaying Ye, Shengyao Liu, Zachary Englhardt, Girish Narayanswamy, Vikram Iyer, Qiuyue (Shirley) Xue

2. **Embedded Arena: Iterative Optimization via Hardware Feedback**
   Conference on Neural Information Processing Systems (NeurIPS), 2026 (Spotlight)
   Zhihan Zhang*, Alexander Le Metzger*, Jiuyang Lyu*, Chun-Cheng Chang, **Jiayi Shao**, Yujia Liu, Emmanuel Azuh Mensah, Edward Wang, Kurtis Heimerl, Gregory D. Abowd, Shwetak Patel, Natasha Jaques, Vikram Iyer

3. **Demo of NailSense: Unobtrusive Wireless Smart Nail Stickers for Whole-Hand Posture Tracking**
   Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (UIST Adjunct '26), 2026 (Demo)
   Aarav Wadhwani, **Jiayi Shao**, Rishabh L. Goenka, Zhenyang Fu, Xiyuan Shen, Qiuyue (Shirley) Xue

4. **Tutorial: Fabricating Electronic Yarns from Off-the-Shelf Circuits for Smart Fabric**
   UbiComp Companion '26, 2026 (Tutorial)
   Jiayi Shao, Chun-Cheng Chang, Zhechen Zhao, Yiyue Luo, Shwetak Patel, Vikram Iyer, Zhihan Zhang, Tingyu Cheng

Awards & Honors

- [2023] Shanghai Outstanding Graduate Award, City of Shanghai (highest honor for graduates in Shanghai)
- [2023] Outstanding Capstone Project Honor, Shanghai University (top 5%)
- [2022] National Students' Innovation and Entrepreneurship Program, National-Level Project
- [2021] 2nd Prize, TI Cup National Undergraduate Electronic Design Contest (1st Prize in Shanghai; 17,313 teams nationwide)
- [2021–2022] University Academic Scholarship, Shanghai University (1st Prize, top 5%; 2nd Prize, top 10%)
- [2019–2022] University Innovation and Entrepreneurship Scholarship, Shanghai University (awarded 3 times)

Technical Skills

Hardware & Embedded Systems: PCB/FPCB design, low-power embedded systems, nRF52, ESP32, BLE, LoRa, RF energy harvesting
Programming & Tools: C/C++, Python, Zephyr RTOS, Arduino, KiCad, J-Link
Sensing & Data Processing: Physiological signal processing, multi-sensor synchronization, power characterization
On-Device ML: Model fine-tuning, embedded deployment, accuracy and robustness benchmarking

Relevant Coursework: Foundations of Machine Learning, Introduction to Artificial Intelligence, Advanced Embedded Systems Design